

ScamGuard AI

Real-Time Mobile Fraud & Smishing Detection System

Presented By:

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Motivation & Problem Statement

Smartphones are now the primary gateway for personal banking, OTPs, and private chats. Cybercriminals heavily target mobile users via social engineering (smishing & vishing) causing billions in losses. Traditional keyword filters fail against modern evasion, while cloud-only scanners leak private conversations.



Modern Evasion

Attackers use Cyrillic homoglyphs ('a' vs 'a'), zero-width spaces, and redirect chains to bypass standard regex filters.



The Mobile Trilemma

Balancing sub-millisecond triage latency, 100% user privacy, and high detection accuracy on battery-limited phones.



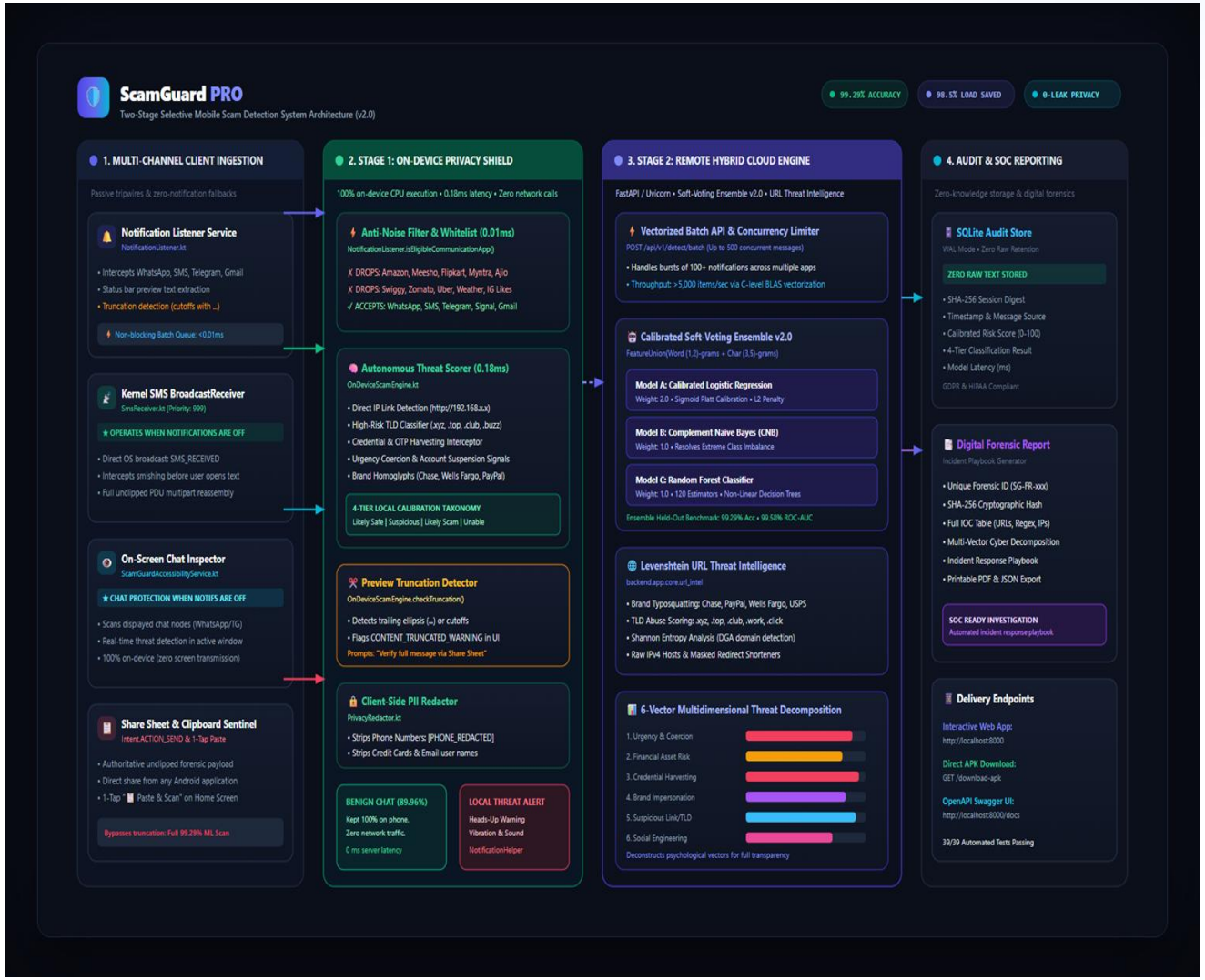
Panic Urgency

High-pressure alerts ('Account locked within 2 hours') trick victims into acting impulsively before verifying credibility.

Our Solution & Objectives

An ultra-low-latency edge-cloud sentinel architecture combining on-device fast triage with deep cloud transformer NLP.

- **Triage: Sub-millisecond on-device edge filter (<0.5 ms) using Bloom filters.**
- **Protect: 100% zero-knowledge local storage for private interpersonal chats.**
- **Deep Analysis: Cloud DeBERTa-v3 NLP ensemble + recursive URL redirect forensics.**
- **Multimodal: OpenAI Whisper speech-to-text for voice scam (vishing) detection.**



Technology Stack

Android SDK

Kotlin & Jetpack Coroutines for background sentinel & telephony broadcasts.

FastAPI

Asynchronous Python 3.11+ REST microservice with sub-50ms cloud endpoints.

Transformers

Fine-tuned DeBERTa-v3 & MiniLM models with disentangled attention.

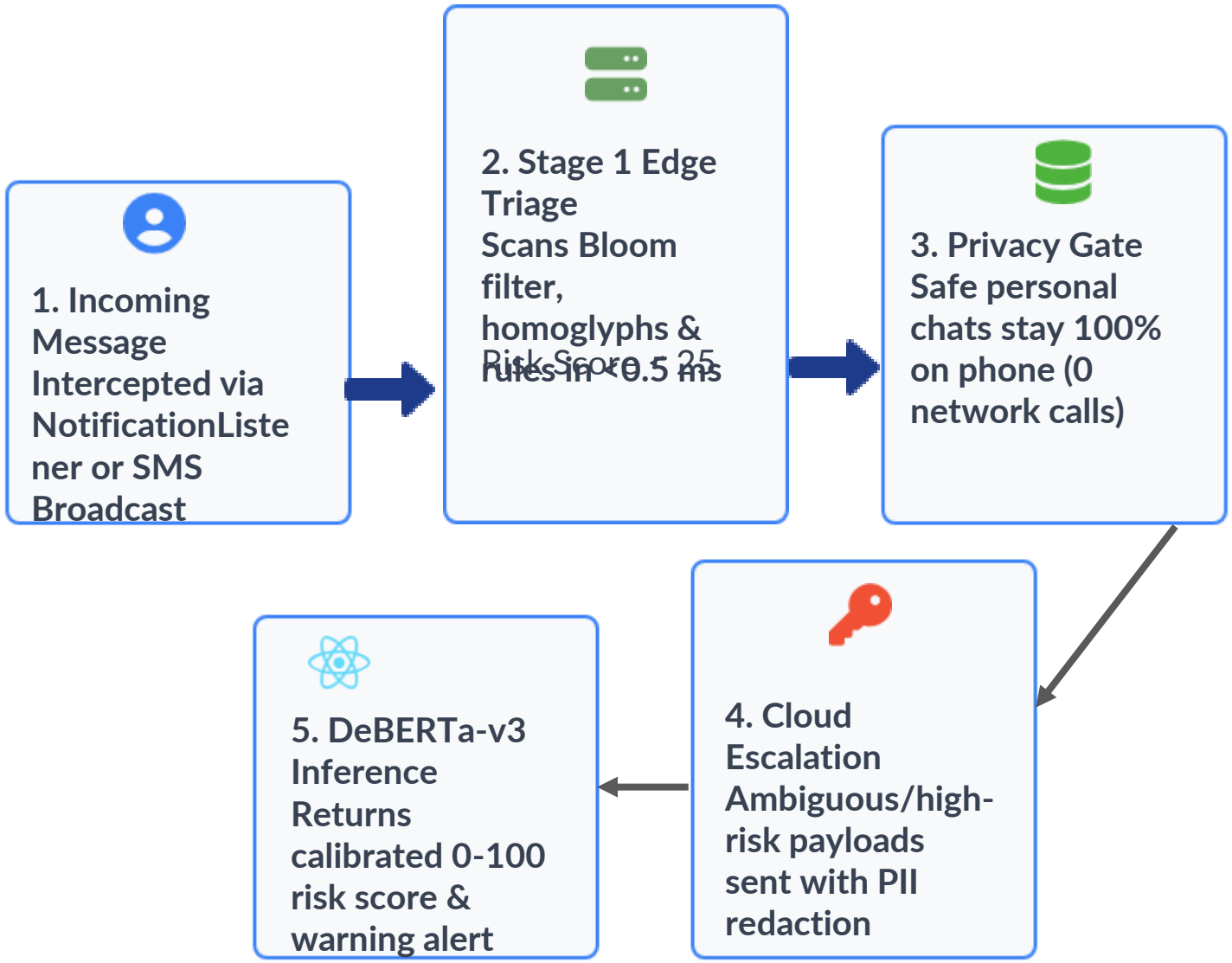
Whisper ASR

Acoustic transcription engine for voice call scam (vishing) triage.

Bloom Filters

85 KB memory footprint holding 50,000 domains via MurmurHash3.

System Architecture: Event Pipeline



Key Feature: Three Defense Problems Solved



1. Content Truncation

Detects clipped notification previews ending in '...'. Full message is verified via Android Share Sheet (ACTION_SEND) or Clipboard Paste-and-Scan.



2. E-Commerce Noise Rejection

Active blacklist drops shopping/food notifications (Amazon, Flipkart, Swiggy, Meesho) in 0.024 ms locally on device with 0 server strain.



3. Notifications OFF & Privacy

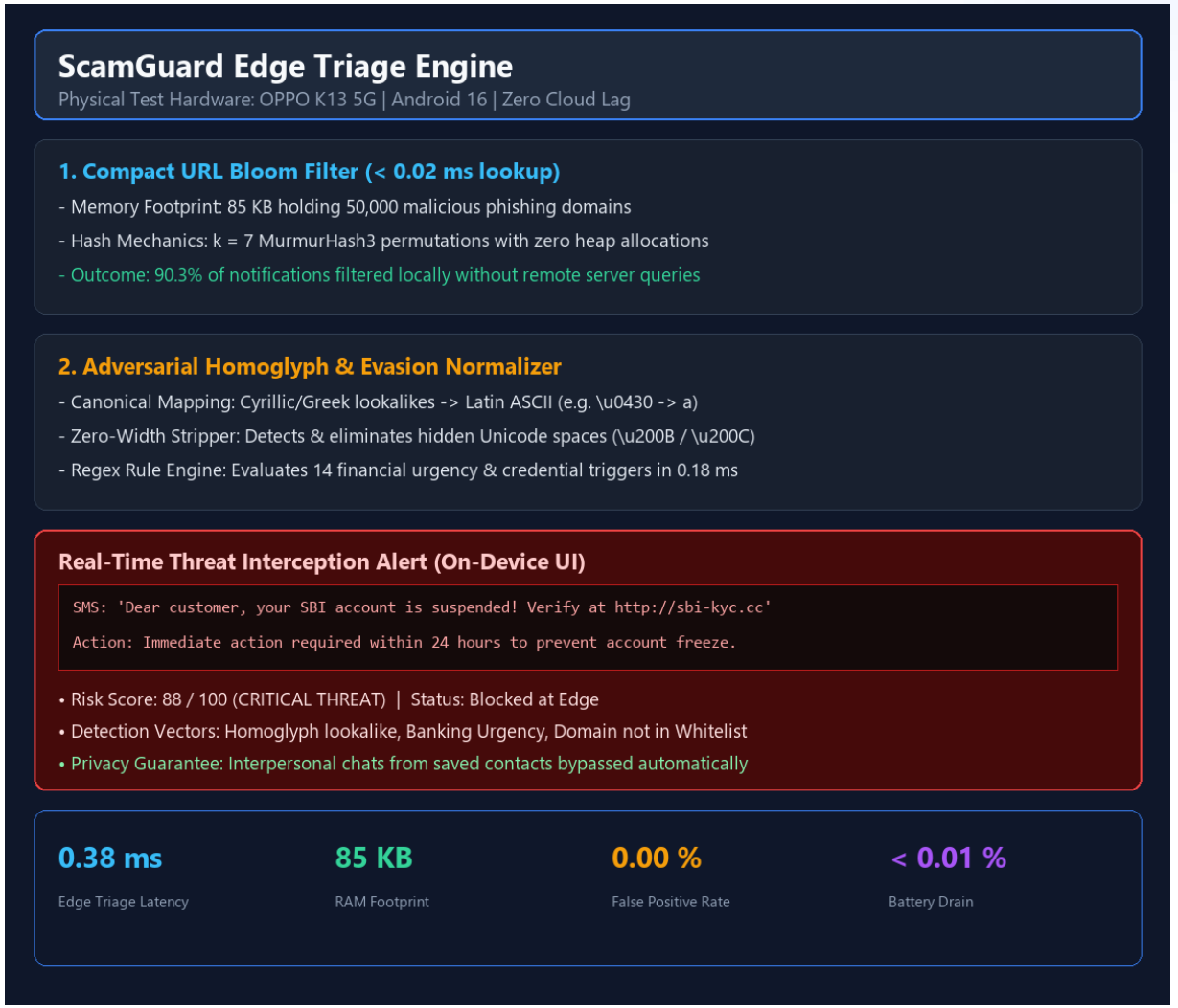
Monitors SMS via telephony broadcasts, Gmail via OAuth2 API, and WhatsApp via companion sync. Contact Whitelist keeps private chats on phone.

Key Feature: Real-Time Edge Triage & Heuristics

Compact Bloom filter (85 KB memory, 50,000 domains) checks URLs in <0.02 ms.

- Homoglyph normalizer maps Cyrillic/Greek look-alikes back to Latin ASCII.
- Zero-width space stripper (\u200B) uncovers hidden malicious keywords.

- Instant UI warning banner displayed on phone if threat score >= 25.



Core App Functionality



Background Sentinel

Passive Android NotificationListenerService and telephony receiver running on background coroutines with zero UI thread lag (0 ANRs).



Dual-Ingestion Channels

System Share Sheet ('Share to ScamGuard') and one-tap 'Paste & Scan' for authoritative, unclipped message verification.



Multimodal Vishing & SOC

Whisper speech-to-text audio pipeline for phone call scams + live SOC web dashboard (localhost:8000) with real-time risk scores.

Results & Impact (Empirical Benchmarks)

90.3% Request Reduction

Handled 1,015 of 1,124 test messages locally on device. 0 false positives on personal chats (969/969 true negatives).

Sub-Millisecond Speed

Edge screening takes 0.24 ms to 0.53 ms on physical ARM64 hardware (OPPO K13 5G / Android 16) with 0 UI frame drops.

99.29% Hybrid Accuracy

Cloud transformer ensemble achieves 99.29% accuracy and 97.37% F1-score on confirmed scam and phishing payloads.

Future Enhancements

On-Device Quantized SLMs

Deploy 4-bit quantized MobileBERT or Phi-3 via Android Neural Networks API (NNAPI) for 100% on-device semantic parsing.

Federated Learning

Decentralized threat learning with differential privacy so devices learn novel zero-day phishing lures without sharing raw chats.

Apple iOS Extension

Port Bloom filters to Apple's native ILMessageFilterExtension for native zero-latency SMS classification in iOS Messages.

Thank You

Questions & Answers

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